Tracking Ticks Across a Changing Climate: A Multi-NEON Site Analysis of Temperature, Humidity, Precipitation on Tick Population Dynamics in the U.S.

Shobha Ratnam Priya, Shaarav Bhosale, Natalie Bowman, Kaitlyn Harrison, and Tyler Le

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Introduction

Recent studies suggest that the range of many tick species is expanding, and that increasing temperatures as well as other changes in climate, such as humidity or precipitation, may be influencing their spread. Ticks have been shown to prefer an environment that is warm, humid, and thickly forested, because it is beneficial to their lifestyle (Soneshine, 2018). They have three stages of life: larvae, nymph, and adult. Ticks are most abundant in their nymph stage, and their populations decline as they grow older. A warm environment helps ticks grow past the nymph and larvae stages and into an adult, and a thick forest has more animals or hosts a tick could latch onto and feed from, leading it to reproduce (Rochlin & Toledo, 2020).

The expansion of tick populations is not just an ecological concern but also a threat to public health. As ticks expand to new areas, they bring their diseases with them, such as Lyme Disease, Alpha-Gal syndrome (which makes you allergic to red meat) and Rocky Mountain spotted fever (Soneshine, 2018 ; Eisen et. al, 2017).

For this study, we looked at data from three NEON sites located in different areas of the United States. Harvard Forest (HARV), located in the Northeast near Boston, Massachusetts, has an average annual temperature of 7.4°C is composed of deciduous forest, evergreen forest, mixed forest, and woody wetlands. University of Notre Dame Environmental Research Center (UNDE) is in the Midwest along the border of Wisconsin and the Upper Peninsula of Michigan and has an average annual temperature of 4.3°C, with vegetation including deciduous forest, mixed forest, and woody wetlands. Oak Ridge (ORNL) is located in the Cumberland Plateau in Tennessee, with an average annual temperature of 14.4°C and composed of deciduous forest, evergreen forest, and pasture. Tick populations are already prevalent in the Northeastern United States, and comparing data from a site located there to sites located further west and south allowed us to determine whether there is evidence that ticks may be spreading in these directions due to climate change. In order to achieve this, we compared the climate measurements from each of these sites with their tick counts to find out if there were any correlations between the two from 2014 to 2024.

Though climate change encourages the expansion of ticks across the United States, there are also factors that limit this expansion. These can include the availability of a tick's preferred host, the type of vegetation and habitat in the area, and human activity. Human activity in particular makes it difficult to accurately predict tick population and expansion because of the increasing amount of land converted to agricultural usage (Sonenshine, 2018).

Problem Statement

This study operates under the framework that climate change is a key driver of vector migration and disease risk, using Lyme disease carrying ticks as a case study. Specifically, it explores the relationship between three climatic variables, temperature, humidity, and precipitation, and tick population (adult and nymph) dynamics across multiple NEON sites in the United States.

We hypothesize that rising temperatures, elevated humidity, and shifting precipitation patterns create more favorable conditions for ticks to thrive and expand geographically, especially in the Eastern U.S., where climate suitability is already high. To test this, we applied a comparative ecological modeling approach, analyzing tick and climate data from 2014 to 2024 across three geographically distinct sites: HARV, UNDE, and ORNL.

Hypothesis

We hypothesize that factors like rising temperatures, increased humidity, and changing precipitation patterns caused by climate change are contributing to the geographic expansion and population growth of ticks in the United States from 2014-2024.

Methodology

Our analysis integrates ecological and climate data to investigate how tick populations vary in response to environmental changes across NEON sites. The methodology followed five key stages: data loading and transformation, merging tick and climate datasets, aggregating tick counts and weather metrics, NEON sites segregation, and visualization for exploratory analysis.

We began by loading NEON datasets that included tick collection records and weather variables such as temperature, humidity, and precipitation. To prepare the tick data, we extracted the month and year from collection dates and created a ‘totalTickPopulation’ column by summing the counts of nymph and adult ticks. Next, we merged the tick and climate datasets using common key factors: month, year, and siteID, ensuring that tick data aligned accurately with the corresponding sites and climate observations.

We refined the merged dataset by computing the mean temperature and mean humidity for each site and time period, standardizing the environmental indicators across observations. We finalized the tick population count by excluding larvae counts, as including them would have inflated population estimates inconsistently across sites. Larval hatching rates are highly sensitive to micro environmental conditions and vary significantly, making comparisons across NEON sites less reliable.

Although our dataset initially covered six NEON sites, we narrowed our focus to three representative locations—ORNL (Oak Ridge National Lab), HARV (Harvard Forest), and UNDE (Underwood)—chosen for their data completeness and geographic diversity. In future work, we plan to expand our analysis to include sites in the Midwest and Western U.S. to better capture tick movement patterns nationwide.

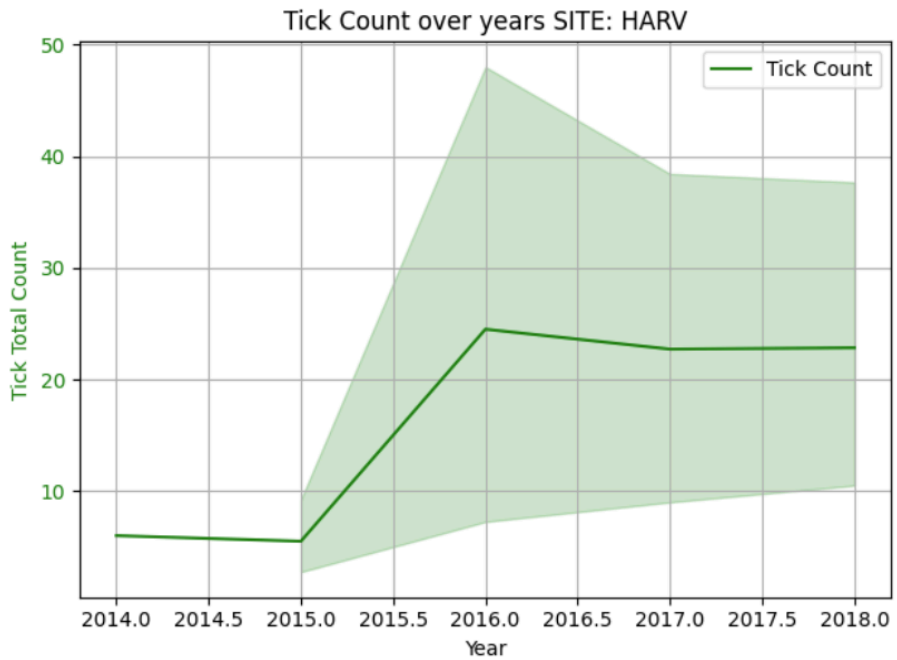
For each selected site, we generated a series of visualizations, including:

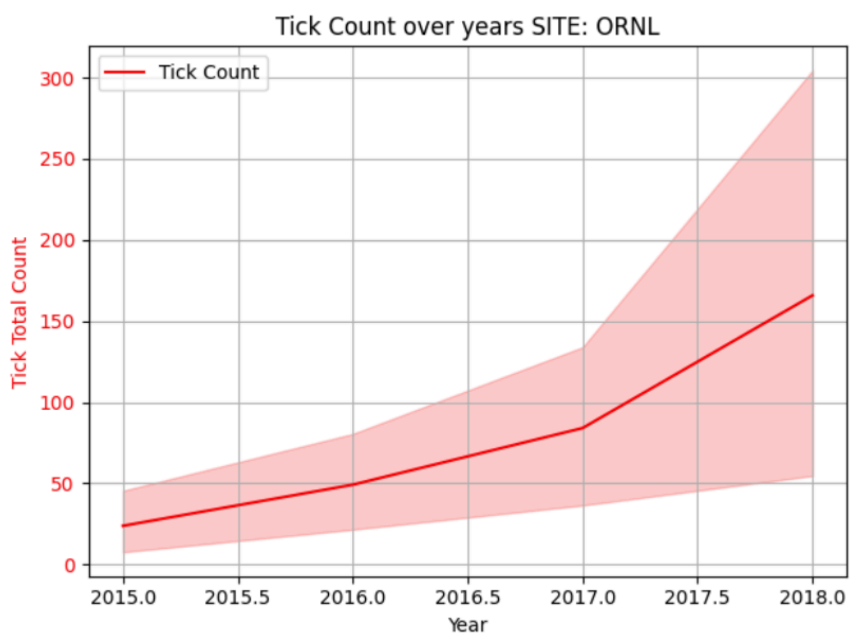
1. Time-series plots of total tick population across months and years.
2. Correlational analysis by measuring the strength of association between tick count and each climate variable
3. Geospatial comparison by evaluating how these relationships differ between NEON sites with distinct climates and ecological zones.

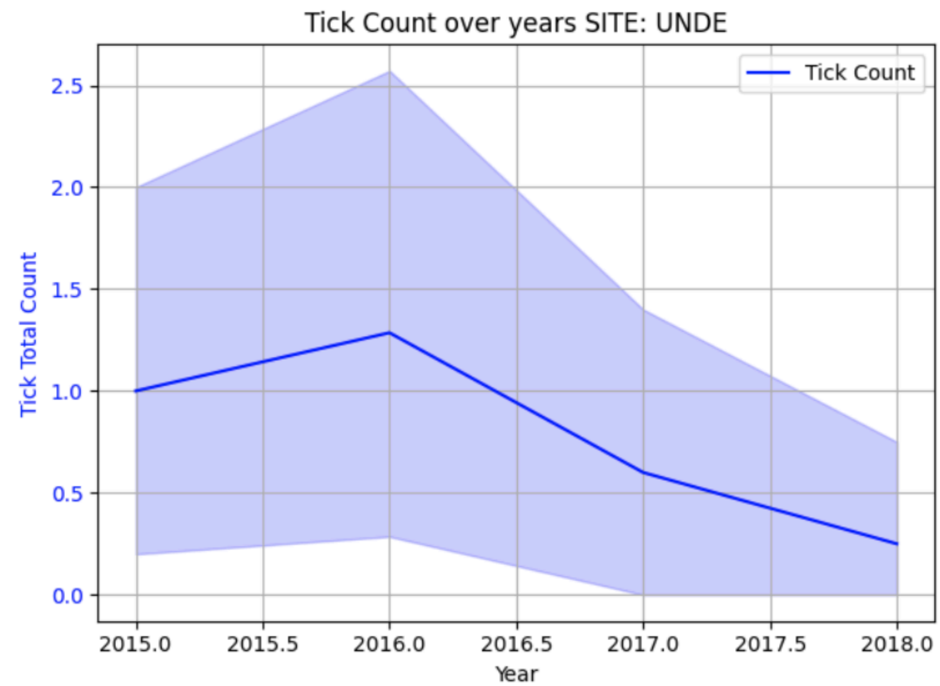
These visual tools enabled us to observe seasonal patterns, investigate regional differences, and identify potential relationships between tick activity and climatic variables.

Results and Discussion

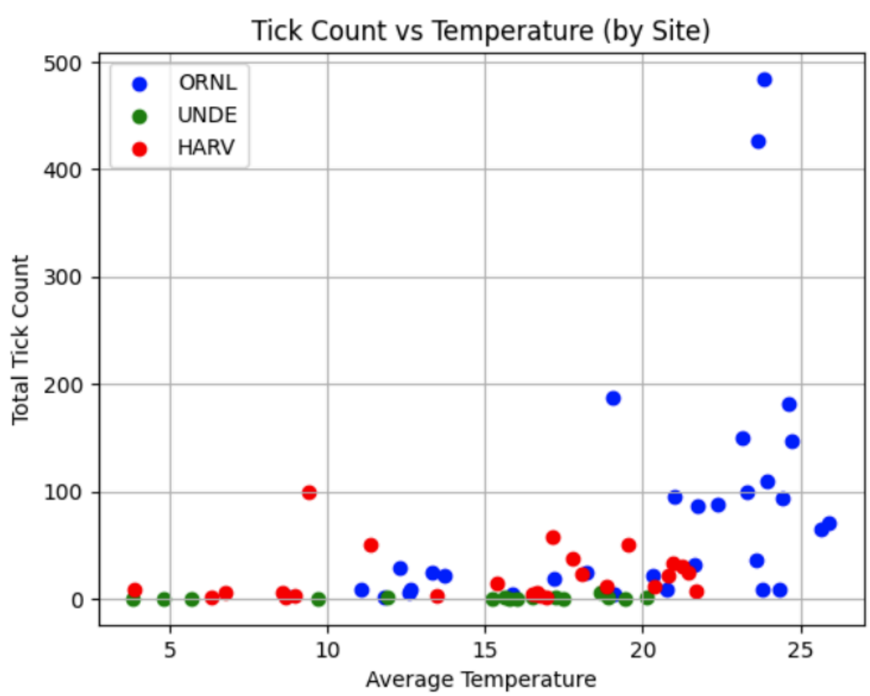
In sites HARV and ORNL, we noticed a net increase in tick populations over the years 2014 to 2018. While the tick population decreased over the years in site UNDE, no conclusions can be drawn from this data as the maximum tick count was 5. Given the small sample size for this site, information collected from UNDE will not be used to present conclusive evidence. It is important to note that total tick count was calculated by adding the adult count and nymph count excluding the larvae count. The total tick count in the ORNL site in Tennessee is greater than the total tick count in the HARV site in Boston.


Tick Count over years SITE: HARV
Tick Count
Tick Total Count
Year
50
40
30
20
10
2014.0
2014.5
2015.0
2015.5
2016.0
2016.5
2017.0
2017.5
2018.0


Tick Count over years SITE: ORNL
Tick Count
Tick Total Count
Year
300
250
200
150
100
50
0
2015.0
2015.5
2016.0
2016.5
2017.0
2017.5
2018.0



The following scatterplot shows the relationship between total tick count and average temperature in celsius. It can be noted that the temperature range between 20 and 25 degrees celsius is most favorable for ticks to thrive in. Tick population increases with an increase in temperature and a greater spray is noticed at higher temperatures.

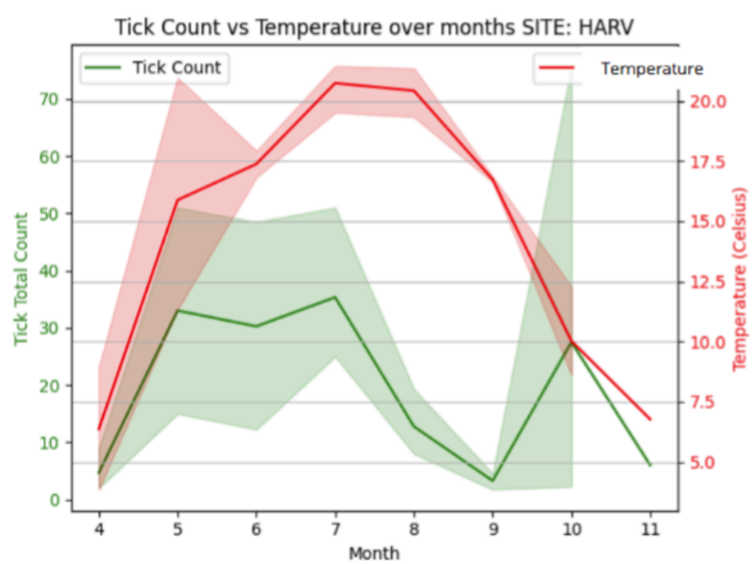


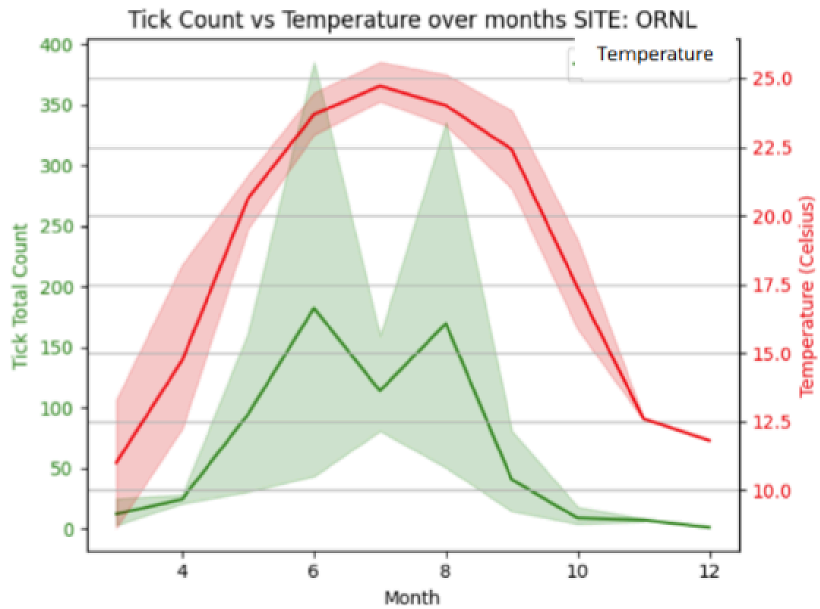
The following graphs show tick counts and temperature trends across months at the HARV and ORNL sites. To evaluate how tick populations change with temperature over time, we analyzed data by month rather than by year. This decision was based on the strong seasonal variations throughout the year. Averaging temperature over an entire year could obscure important seasonal effects—for instance, an unusually harsh winter could lower the annual average temperature, resulting in misleading conclusions about general weather conditions.

To address this, we averaged temperatures by month across multiple years, as certain months consistently tend to be warmer or colder. This method provides a clearer picture of seasonal temperature changes and how ticks respond to them.

From the graphs, it's evident that tick populations increase during the warmer months and drop sharply in colder months. This helps explain why tick populations are higher at ORNL compared to HARV. Tennessee (ORNL) experiences a longer stretch of warm months—specifically months 5 through 9—with temperatures exceeding 20°C. In contrast, Boston (HARV) only reaches similar temperatures during months 7 and 8. Additionally, ORNL reaches higher maximum temperatures overall.

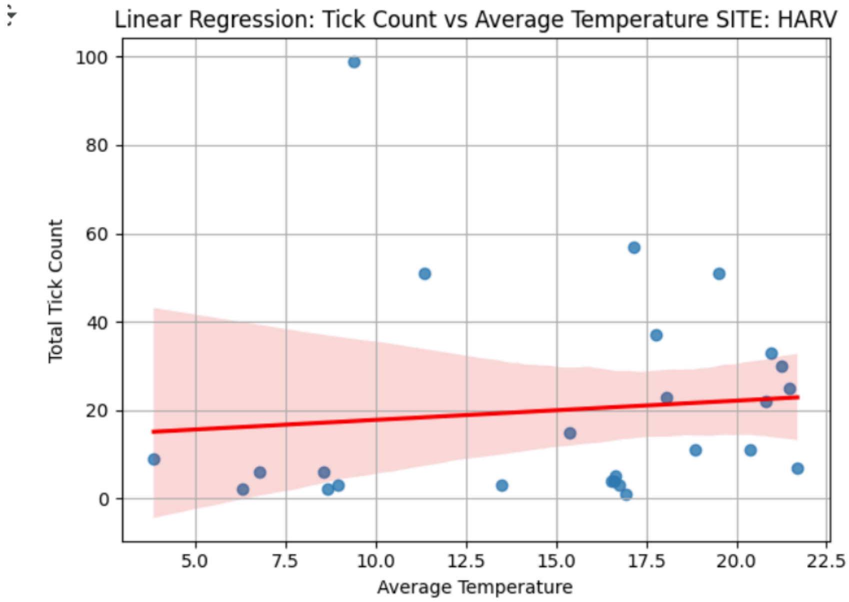
Since it's already been established that ticks prefer warmer conditions, it's easy to see why the warmer site supports a larger tick population than the cooler one.

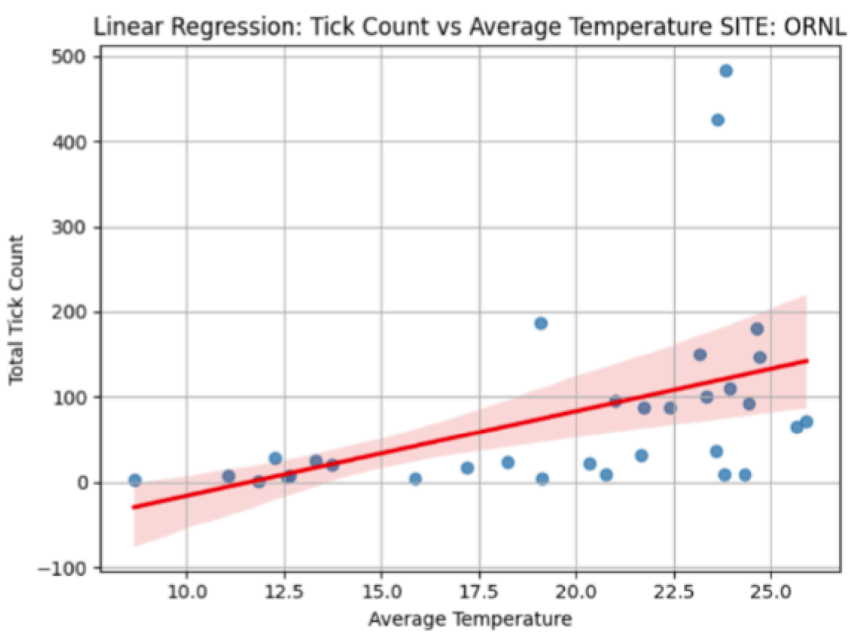




The following graphs show linear regression analyses between average temperature and tick count. While we initially hypothesized a stronger correlation in the eastern U.S. compared to the western U.S., the results suggest otherwise. A stronger correlation is observed at ORNL, which lies to the west of HARV.

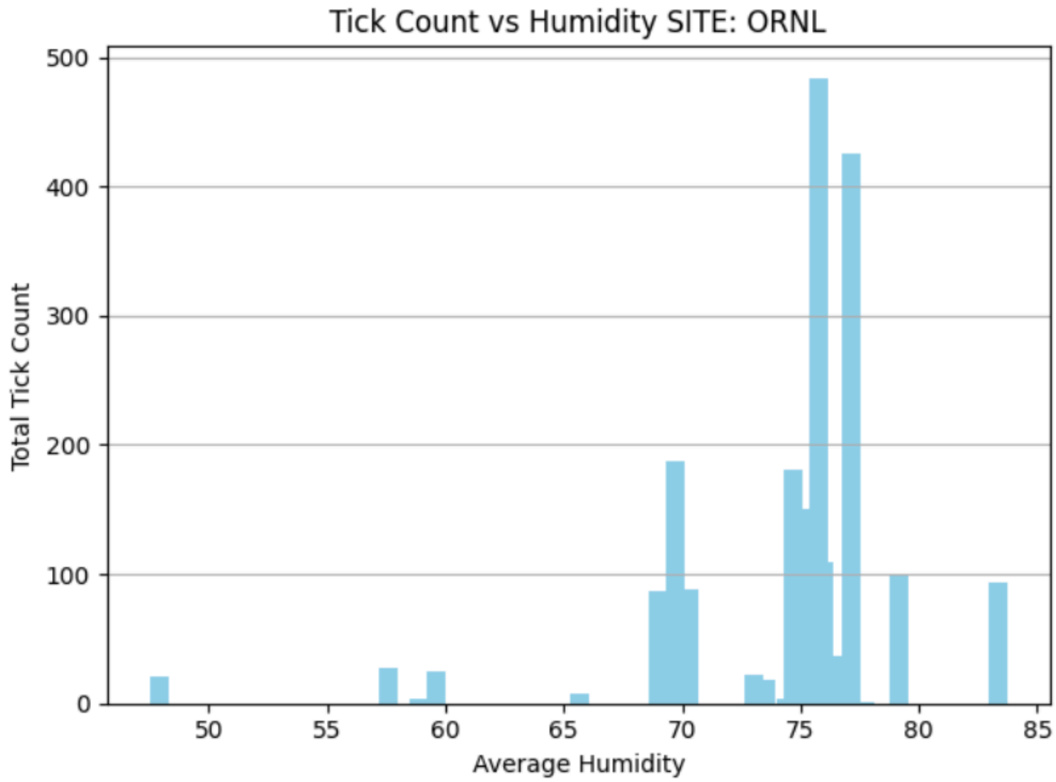
This result can be attributed to several factors. ORNL, located in Tennessee, experiences a longer and more consistent warm season, allowing for a sustained tick presence and a clearer relationship between temperature and tick population. In contrast, HARV in Massachusetts has a shorter window of favorable temperatures and more variable seasonal shifts, which may weaken the temperature-tick correlation. Additionally, Tennessee falls within a known high-risk tick spread region, where environmental conditions—such as humidity, vegetation cover, and host availability—are more favorable for tick survival and reproduction. These combined factors likely contribute to the stronger observed correlation at ORNL.

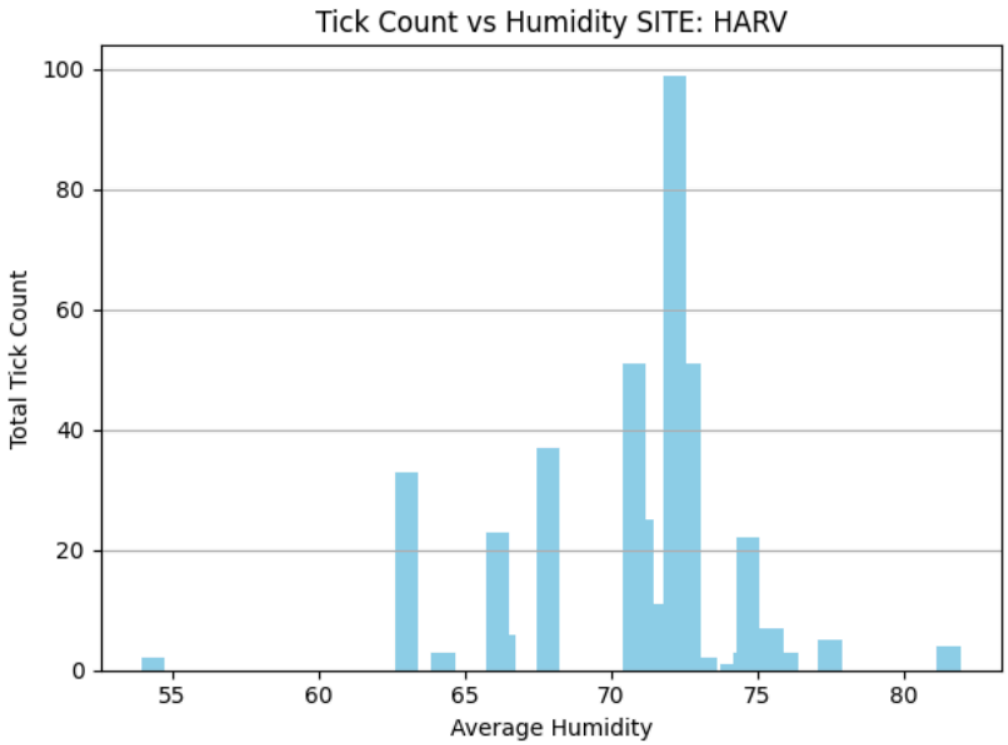


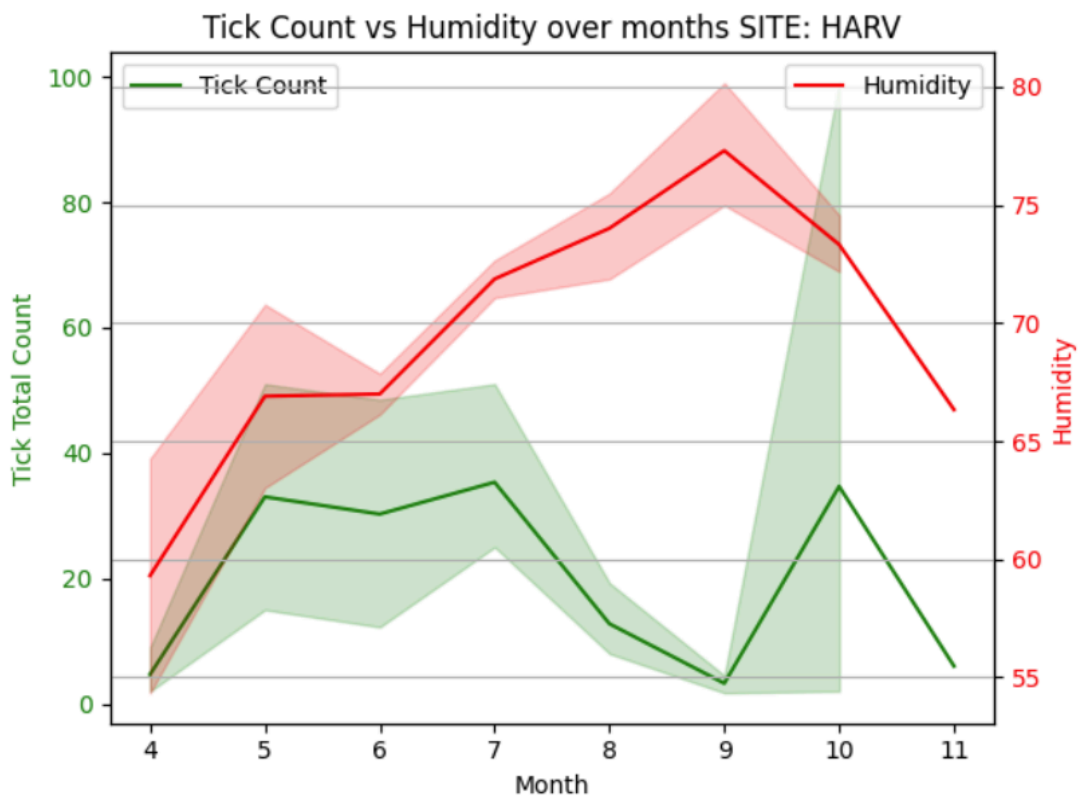


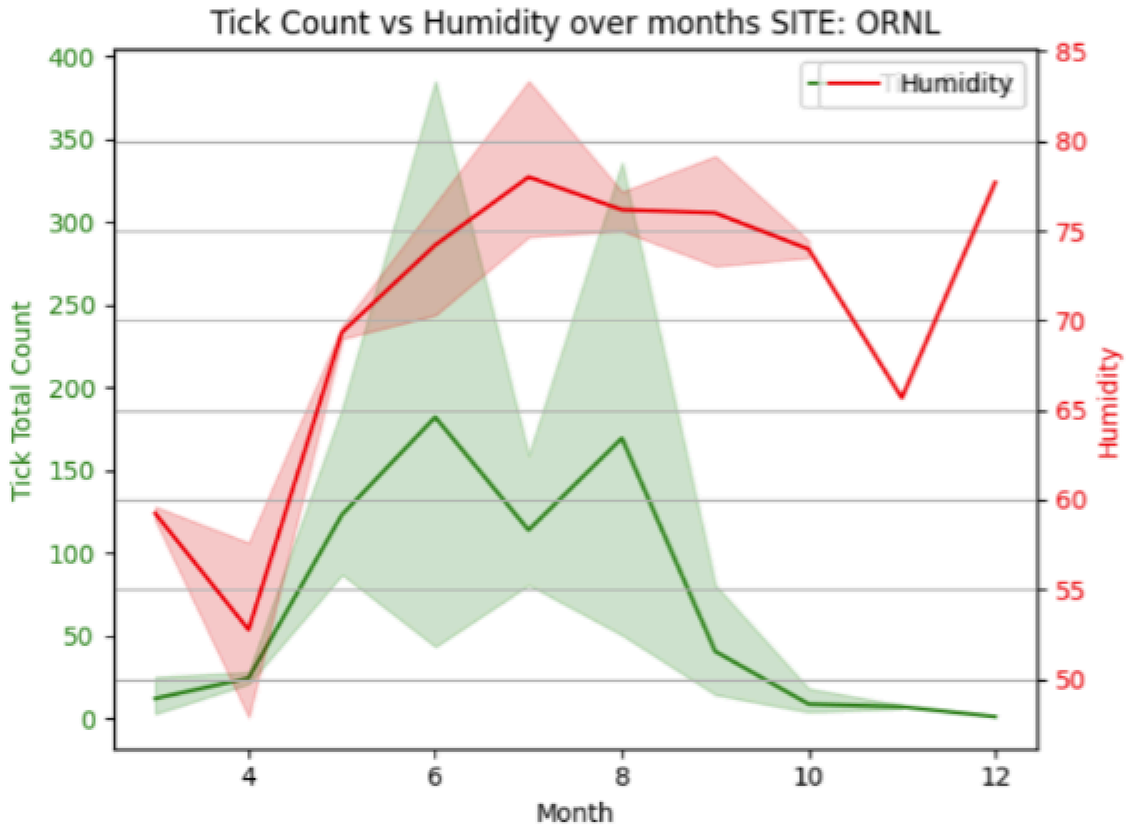
From the following bar graphs showing the relationship between tick count and relative humidity, it is evident that ticks thrive best at RH levels between 70% and 80%. Tick counts decline when humidity falls below or rises above this optimal range. It can also be noted that ticks from the HARV site prefer slightly lower humidity levels (70-75%) than ticks in the ORNL site (75-80%).

This difference in preferred humidity levels may be due to local environmental adaptation. Ticks at the HARV site, located in Massachusetts, are exposed to generally cooler and less humid conditions compared to those at ORNL in Tennessee. Over time, the tick population at HARV may have adapted to thrive at slightly lower humidity levels (70–75%). In contrast, the warmer and more humid climate of ORNL supports ticks that are better suited to slightly higher humidity levels (75–80%). These variations reflect how local climate conditions can shape physiological tolerances in tick populations.

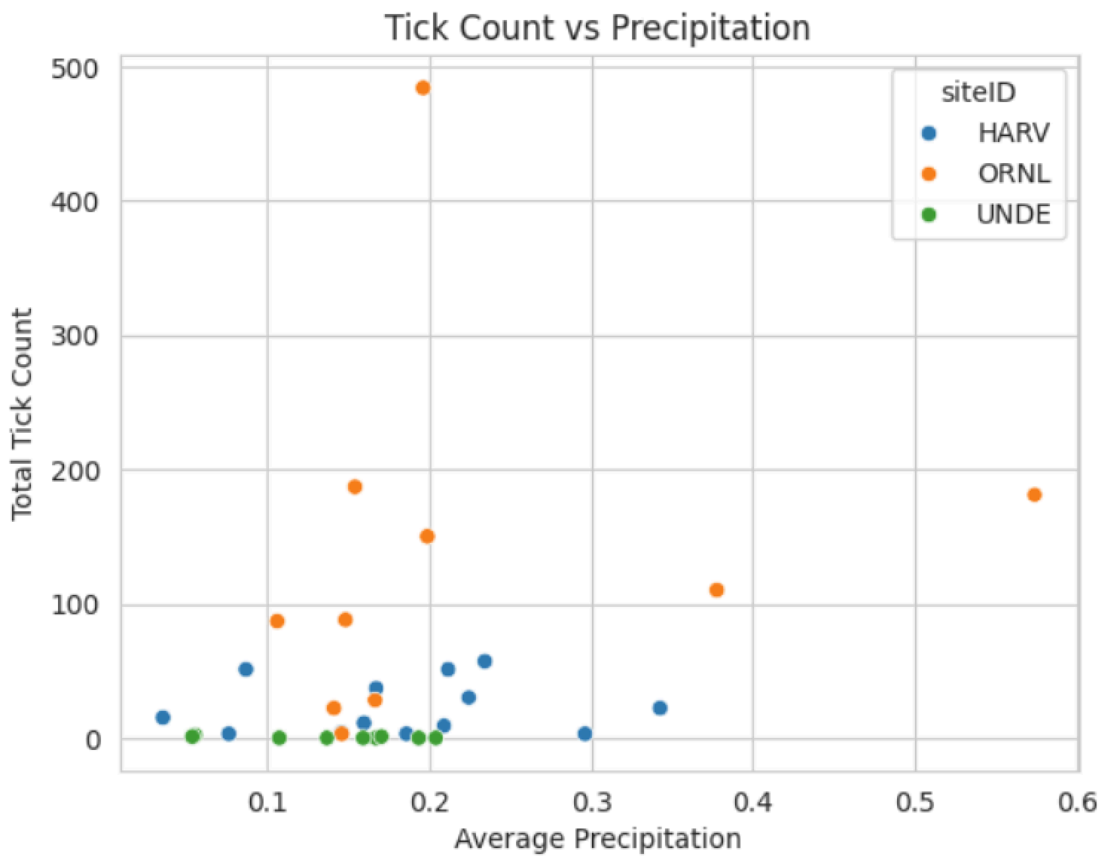



Tick Count vs Humidity SITE: HARV
Total Tick Count
0
20
40
60
80
100
55
60
65
70
75
80
Average Humidity

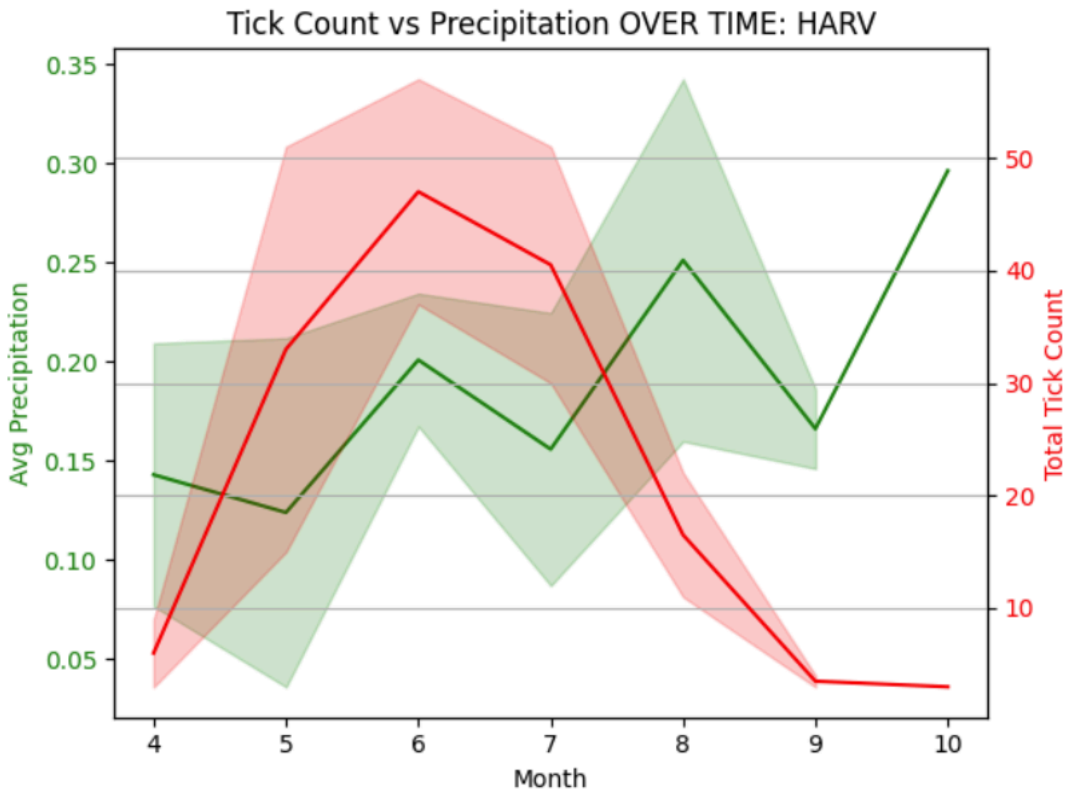

Tick Count vs Humidity over months SITE: HARV
Tick Count
Humidity
Tick Total Count
0
20
40
60
80
100
55
60
65
70
75
80
4
5
6
7
8
9
10
11
Month

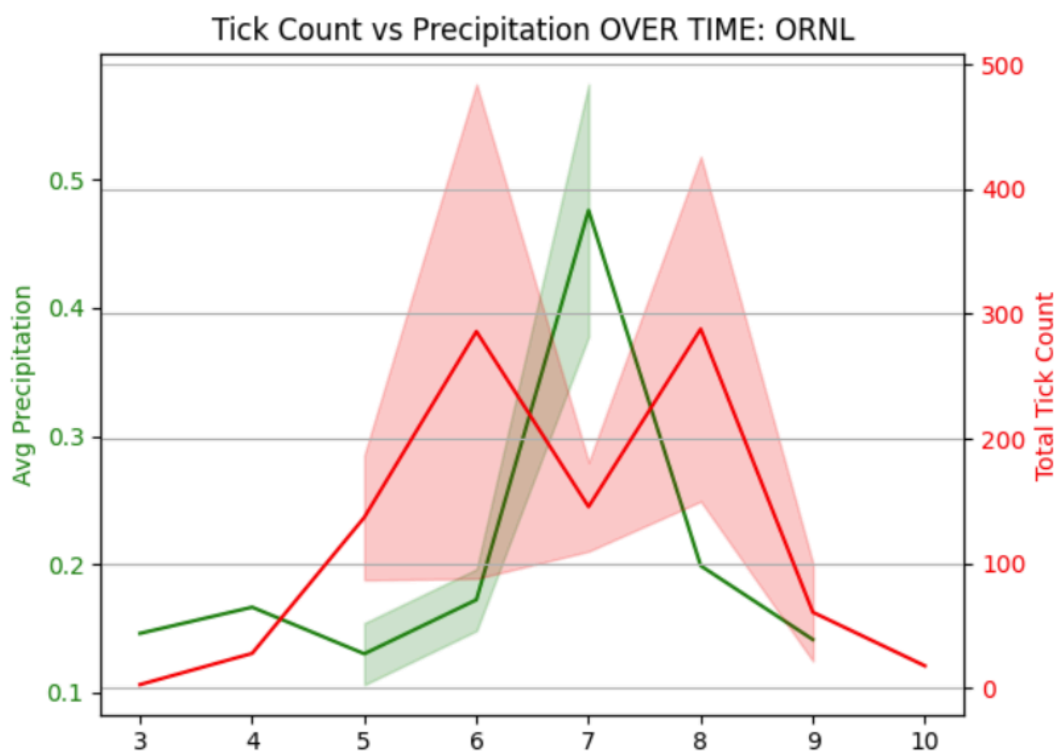


The graph illustrates the relationship between tick count and precipitation in millimeters. It suggests that ticks tend to favor light precipitation levels, as there is a more noticeable spread in tick counts within the 0.1 to 0.2 mm precipitation range.It appears that both sites, HARV and ORNL, experience light precipitation throughout the year, with 0.6 mm being the maximum recorded level.



It is difficult to establish a clear relationship between precipitation levels and tick count, as the HARV site shows no evident correlation, while the ORNL site exhibits only a slight negative correlation. Additionally, both sites receive very light rainfall throughout the year, with precipitation levels remaining low, which limits the ability to draw meaningful conclusions from the available data. Therefore, we turned to existing literature to explore what other studies have found regarding the relationship between precipitation and tick populations.


Tick Count vs Precipitation OVER TIME: HARV
Avg Precipitation
Total Tick Count
Month
0.35
0.30
0.25
0.20
0.15
0.10
0.05
50
40
30
20
10
4
5
6
7
8
9
10


Tick Count vs Precipitation OVER TIME: ORNL
Avg Precipitation
Total Tick Count
0.5
0.4
0.3
0.2
0.1
500
400
300
200
100
0
3
4
5
6
7
8
9
10

Research Context and Conclusion

Previous research has demonstrated significant relationships between climate factors and tick population changes. Sonenshine (2018) established that ticks tend to prefer warm, humid, and thick forested environments, which aid in their life cycle necessities. These findings directly align with our observations that tick populations are most abundant at sites with temperatures ranging from 20 - 25°C and humidity levels of 70-80%.

Ogden et al. (2013) identified a "roughly quadratic relationship between temperature and host seeking activity", highlighting that tick activity is inhibited at both low and high temperature extremes. Their work also noted that "low relative humidity at one extreme, and intense rainfall at the other are inhibitory of host-seeking activity", establishing an important boundary conditions for ticks. Their observations support our findings of minimal tick activity below 10°C and reduced activity during times of heavy precipitation.

The role of precipitation in tick ecology appears to be more nuanced than our data alone can establish. As our analysis indicated, it was difficult to establish a clear relationship between precipitation levels and tick count. Nonetheless, existing literature provides valuable context. Jones and Kitron (2000) and Berger et al. (2014) assert that 'greater precipitation increases leaf litter moisture, thus creating favorable conditions for questing activity and tick survival' (as cited in Mark, 2022). It is important to note that Ogden et al. (2013) recognized that while moderate amounts of moisture benefits ticks, 'intense rainfall' can inhibit host-seeking behavior, suggesting a theoretical boundary condition that our study sites may not have fully experienced. Mark (2022) goes on to explain that rainfall can 'stabilize temperature fluctuations in the microclimate under the soil surface, thus affecting tick development'. This microclimate effect may be present even at the precipitation levels observed in our sites, potentially explaining variations in tick abundance that is not directly apparent in our analysis.

These findings have important public health implications as climate change creates more favorable conditions for ticks to expand. The geographic differences in climate suitability suggest that regions experiencing warming temperatures and maintaining optimal humidity levels may face increased tick abundance and associated disease risks. Future research should incorporate data from additional NEON sites across diverse regions and develop predictive models that might account for complex climate interactions to better forecast tick population trends and implement targeted prevention strategies in vulnerable regions.

References

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